

Hide menu

- Modern Cipher and Module Objectives
- Modern Cipher and Block Cipher (vs. Stream Cipher)
- Ideal Block Cipher
- Feistel Cipher
- Data Encryption Standard (DES)
- Video: DES Overview

3 min
- Video: DES Round Function

1 min
- Reading: About Pseudocode

30 min
- Peer-graded Assignment: DES Overview Pseudo-Code

1h 10m
- Review Your Peers: DES Overview Pseudo-Code
- Video: DES Subkey Generation

1 min
- Video: DES Security

4 min
- Reading: Lecture Slides for Block Cipher and DES

15 min
- Quiz: DES

Submitted

• Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

DES

To pass 80% or higher

Retake the assignment in 7h 59m

Go to next item

Review Learning Objectives

1. True or False: DES is an ideal block cipher.

2 / 2 points

☐ True

☒ False

☒ Correct

See Feistel Cipher Lecture.

Submit your assignment

Due Apr 21, 11:59 PM IST Attempts 3 every 8 hours

Try again

Retake the quiz in 7h 59m

Receive grade

To Pass: 80% or higher

2. True or False: DES displaying Avalanche Effect is a limitation because it describes that an error occurring in one of the rounds propagate through the rest of the rounds.

Your grade 100%
2 / 2 points

View Feedback

We keep your highest score

☐ True

☒ False

☒ Correct

See DES Security Lecture.

Like

Dislike

Report an issue

3. An attacker is equipped with a computer that performs 10 trillion (10^{13}) DES decryptions per second, what is the average time required, in hours, for a brute force attacker to break DES?

4 / 4 points

1

☒ Correct

DES uses a 56-bit key

4. Which of the followings are true about Feistel Cipher? Select all that applies.

4 / 4 points

☒ Feistel Cipher is a product cipher.

☒ Correct

See Fiestel Cipher lesson.

☒ Feistel Cipher processes the data in halves.

☒ Correct

See Fiestel Cipher lesson.

☒ Feistel Cipher requires smaller key than ideal block cipher.

☒ Correct

See Fiestel Cipher lesson.

☐ Feistel Cipher competed with DES and got outdated after the wide use of DES.

☐ The subkeys used in the Feistel Cipher rounds are independent to each other.

☐ Feistel Cipher requires different encryption and decryption implementations in hardware and software.